Unit-1

Introduction to Cyber Law and Cyber Crime:-

Definition of cyber law, History of Cyber-crime, Types of cyber-crimes, classification of cyber-crimes, Distinction between cyber-crimes and conventional crime, Need of cyber law, cyber criminals, cyber crime in India.

Unit-2

Information Technology Act 2000 Introduction to IT Act 2000:-

Objective of the IT Act 2000,structure of the Act, features of IT Act 2000,Important chapters of IT Act, summary of the Act, Amendment bill 2008

Unit-3

Cyber Crime Investigation:-

Importance of cyber-crime investigation, Motive behind cyber-crime, steps and procedures during investigation, Procedure for search and seizure of digital evidence

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International Regime in cyber laws and Intellectual Property Rights :-

Introduction United nations and other international organization,IPR in India,Patents,Tradmarks,Industrial designs, and Geographic indication of source

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Electronic Business and legal issues in IT Act :-

Evolution and development in E-commerce, paper vs. paper less contracts E-commerce models

Unit-6

Cyber Crime and Other Laws :-

Indian Evidence Act and cyber-crimes, Indian Penal code and cyber-crimes, criminal Procedure code and cyber crimes

Unit-7

Practical Study Cyber Crime Laws:

Harassment Via E-mails, Email spoofing, Cyber pornography, Cyber Stalking.

Unit-1

Introduction to Cyber Law and Cyber Crime:-

Definition of cyber law, History of Cyber-crime, Types of cyber-crimes, classification of cyber-crimes, Distinction between cyber-crimes and conventional crime, Need of cyber law, cyber criminals, cyber crime in India.

Introduction to Cyber Law and Cyber Crime

Introduction to Cyber Law and Cyber Crime

Introduction to Cyber Law

In the modern digital era, computers, smartphones, cloud computing, and the Internet have become an essential part of our daily lives. We use digital technologies for banking, education, healthcare, e-commerce, social networking, communication, and government services. While these technologies have made life easier and more efficient, they have also created new opportunities for cyber criminals to commit crimes in the digital world.

To regulate these online activities and protect individuals, organizations, and governments from cyber threats, a specialized branch of law called **Cyber Law** has been developed.

Cyber Law is also known as **Internet Law**, **Cyber Space Law**, or **Information Technology Law**. It provides legal rules for the use of computers, digital devices, communication networks, and the Internet. Cyber Law ensures that electronic transactions are legally recognized and that cyber crimes are investigated and punished according to the law.

Cyber Law is not limited to cyber crimes. It also deals with electronic contracts (E-Contracts), digital signatures, intellectual property rights, privacy protection, data security, electronic evidence, e-commerce, online banking, and social media regulations.

In India, the **Information Technology (IT) Act, 2000** is the primary legislation governing cyber law. It provides legal recognition to electronic records and digital signatures and defines various cyber offences and penalties.

Definition of Cyber Law

Cyber Law is the branch of law that governs the use of computers, digital devices, communication networks, and the Internet. It provides legal protection against cyber crimes and regulates electronic transactions, digital communications, privacy, and information security.

Real-Life Example (Cyber Law)

Example 1: Online Banking Fraud

Suppose a criminal creates a fake banking website that looks exactly like the official website. A customer enters their User ID, Password, and OTP on the fake website. The criminal steals the banking credentials and transfers money from the victim's account.

In this situation, Cyber Law helps the police investigate the offence, collect digital evidence, identify the criminal, and punish the offender under the Information Technology Act, 2000 and other applicable laws.

Introduction to Cyber Crime

A **Cyber Crime** is any unlawful activity in which a computer, digital device, network, or the Internet is used either as the tool for committing the crime or as the target of the crime.

Cyber crimes can affect individuals, businesses, educational institutions, hospitals, financial organizations, and governments. Unlike conventional crimes, cyber crimes can be committed from anywhere in the world using digital technologies.

With the rapid growth of online banking, cloud computing, mobile applications, artificial intelligence, and social media, cyber crimes have become more sophisticated and difficult to detect.

Definition of Cyber Crime

Cyber Crime is any illegal activity committed using computers, digital devices, communication networks, or the Internet to steal information, cause damage, commit fraud, or disrupt digital services.

Examples of Cyber Crime

Example 1: Phishing

A cyber criminal sends a fake email claiming to be from a bank and asks the user to verify their account. The victim clicks the fake link and enters banking credentials, which are stolen.

Example 2: Ransomware

A company receives an infected email attachment. After opening it, all files become encrypted, and the attacker demands money to unlock them.

Example 3: Social Media Account Hacking

An attacker gains unauthorized access to a person's Instagram or Facebook account and posts fake content or demands money from friends.

Difference between Cyber Law and Cyber Crime

Cyber Law	Cyber Crime
A legal framework governing cyberspace.	An illegal activity committed in cyberspace.
Protects digital users and organizations.	Causes harm to users, organizations, or governments.
Defines legal rights, duties, and punishments.	Violates laws and regulations.
Focuses on prevention and regulation.	Focuses on illegal acts.
Example: IT Act, 2000	Example: Hacking, Phishing, Ransomware

History of Cyber Crime

Introduction

The history of cyber crime is closely linked with the development of computers and the Internet. As technology evolved from standalone computers to globally connected networks, criminals also changed their methods. Earlier crimes required physical presence, but today a cyber criminal can attack a victim sitting thousands of kilometers away.

Cyber crime has evolved from simple unauthorized access to highly organized attacks involving artificial intelligence, ransomware, cryptocurrency fraud, and cyber terrorism.

Evolution of Cyber Crime (Timeline)

1. 1960s – Beginning of Computer Era

During the 1960s, computers were mainly used in government organizations, universities, and research laboratories. They were large mainframe systems with limited users.

At this stage, cyber crime did not exist in its modern form. However, a few cases of unauthorized access and misuse of computer systems were reported.

Example

A student secretly uses another user's computer account without permission.

2. 1970s – First Network Attacks

During the 1970s, computer networks started expanding. The first experiments with computer networking (ARPANET) connected computers over long distances.

This period saw some of the earliest hacking attempts and password theft incidents.

Example

Someone steals another user's password and logs into the system.

3. 1980s – Birth of Computer Viruses

The 1980s marked the beginning of computer viruses and malware.

One of the first famous computer viruses was **Brain Virus (1986)**, developed by two Pakistani brothers. Although it was originally created to prevent software piracy, it spread worldwide.

Other malicious programs such as worms and Trojan horses also appeared.

Example

A virus spreads through floppy disks and infects hundreds of computers.

4. 1990s – Internet Revolution

The rapid growth of the Internet changed the nature of cyber crime.

Criminals started using email, websites, and chat services to commit fraud.

New cyber crimes emerged:

- Hacking, Email fraud, Website defacement, Online scams, Credit card fraud

Example

A hacker changes the homepage of a company's website.

5. 2000–2010 – Rise of E-Commerce and Online Banking

English

As online banking and e-commerce became popular, financial cyber crimes increased rapidly.

India introduced the **Information Technology Act, 2000**, which became the first comprehensive cyber law.

Common crimes included:

- Internet banking fraud
- Phishing
- Identity theft
- Online shopping fraud

Example

A fake banking website steals customers' passwords.

6. 2010–2020 – Social Media and Smartphones

The widespread use of smartphones and social media created new cyber threats.

Criminals started targeting users through Facebook, WhatsApp, Instagram, and Twitter.

Major attacks included:

- Ransomware , Social engineering, Fake news, Mobile banking fraud
- Cryptocurrency fraud

Example

Someone creates a fake WhatsApp message asking users to click a malicious link.

7. 2020 – Present (AI Era)

Today's cyber criminals use Artificial Intelligence, Machine Learning, Cloud Computing, Deepfake Technology, and Cryptocurrency.

Modern cyber crimes include:

- AI-generated phishing emails
- Deepfake videos
- Crypto scams
- Supply chain attacks
- Cloud attacks
- IoT attacks
- Ransomware

Cyber attacks today are faster, more automated, and more difficult to detect.

Example

A deepfake video is created to impersonate a CEO and trick employees into transferring money.

Major Cyber Crime Milestones

Year	Event	Importance
1960s	Mainframe Computer Era	Unauthorized access begins
1970s	ARPANET	First network attacks
1986	Brain Virus	First famous PC virus
1988	Morris Worm	One of the first Internet worms
1990s	Internet Growth	Hacking and online fraud increase
2000	IT Act, 2000 (India)	Legal framework for cyber law
2017	WannaCry Ransomware	Global ransomware attack
2020+	AI & Deepfake	New generation cyber threats

History of Cyber Crime in India

Cyber crime in India began increasing after the introduction of the Internet in the mid-1990s. The rapid adoption of online banking, digital payments, smartphones, and e-commerce increased cyber risks.

To address these issues, India enacted the **Information Technology Act, 2000**, established **CERT-In**(Indian Computer Emergency Response Team)) and launched the **National Cyber Crime Reporting Portal** to strengthen cyber security.

Types of Cyber Crimes

Introduction to Cyber Crimes

A **Cyber Crime** is any illegal activity in which a computer, mobile device, network, or the Internet is used either as a tool to commit a crime or as the target of the crime. Cyber crimes affect individuals, organizations, businesses, financial institutions, and governments. Due to the rapid growth of digital technology, cyber crimes have become more sophisticated, making cyber security an essential requirement.

1. Hacking

Definition

English: Hacking is the unauthorized access to a computer system, network, or digital device.

How it Works

- Attacker finds a security weakness.
- Gains unauthorized access.
- Steals or modifies information.

Example

A hacker gains access to a university server and changes students' examination results.

Prevention

- Strong Password
- Multi-Factor Authentication
- Firewall
- Antivirus
- Regular Software Updates

2. Phishing

Definition

Phishing is a cyber attack where fake emails, SMS, or websites are used to steal sensitive information.

Example

A fake SBI website asks users to enter Internet Banking credentials.

Prevention

- Check website URL.
- Never share OTP.
- Avoid suspicious links.

3. Identity Theft

Definition

Identity Theft is the unauthorized use of another person's personal information for fraud.

Example

Using someone's Aadhaar card to obtain a fake loan.

Prevention

- Protect personal information.
- Enable two-factor authentication.
- Monitor bank transactions.

4. Malware

Definition

Malware means **Malicious Software** designed to damage computers or steal information.

Types of Malware

1.Virus

Attaches itself to a file and spreads when the file is executed.

Example: USB drive infects multiple computers.

2.Worm

A worm spreads automatically through networks without user action.

Example: Morris Worm.

3.Trojan Horse

A malicious program disguised as legitimate software.

Example: Fake antivirus software.

4.Ransomware

Encrypts files and demands ransom money.

Example: WannaCry Ransomware Attack (2017).

Prevention

- Antivirus software
- Backup important files
- Avoid unknown downloads

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5 Cyber Stalking

Cyber Stalking refers to repeatedly harassing or threatening someone using the Internet or social media.

Example

Sending threatening messages through Instagram or WhatsApp.

Prevention

- Block unknown users.
- Report abusive accounts.
- Keep social media accounts private.

6. Cyber Bullying

Cyber Bullying is the use of digital platforms to insult, threaten, or humiliate others.

Example

Posting insulting comments or fake images on social media.

Prevention

- Report abusive content.
- Do not respond to online harassment.
- Inform parents or authorities.

7. Email Spoofing

Email Spoofing is sending emails using a fake sender address.

Example

An email pretending to be from a bank asking for account details.

Prevention

- Verify sender email.

- Avoid clicking suspicious attachments.

8. DoS and DDoS Attack

Definition

A Denial of Service (DoS) attack floods a server with excessive requests, making it unavailable.

A Distributed DoS (DDoS) attack uses thousands of infected computers (Botnet) to attack simultaneously.

Example

An online shopping website becomes unavailable during a festive sale because of a DDoS attack.

Prevention

- Firewalls
- Intrusion Detection Systems
- Traffic Filtering

9. Online Banking and UPI Fraud

Criminals use fake payment links, QR codes, OTP scams, and fake banking apps to steal money.

Example

A victim scans a fake QR code and loses ₹20,000.

Prevention

- Never share OTP.
- Verify payment requests.
- Use official banking applications.

10. Cyber Terrorism

Cyber Terrorism refers to cyber attacks against government systems, military networks, or critical infrastructure with the aim of creating fear or disrupting national security.

Example

Attacking a power grid or government website.

Prevention

- Strong national cyber security policies
- Regular security audits
- Network monitoring

Summary Table

Cyber Crime	Main Target	Example
Hacking	Computer System	Server Hacking
Phishing	User Information	Fake Bank Website
Identity Theft	Personal Identity	Aadhaar Misuse
Malware	Computer	Virus, Worm
Ransomware	Files	WannaCry
Cyber Stalking	Individual	Online Harassment
Cyber Bullying	Individual	Social Media Abuse
Email Spoofing	Email Users	Fake Bank Email
DoS/DDoS	Website	Server Crash
Banking Fraud	Bank Customers	UPI Scam
Cyber Terrorism	Government	Attack on Government Network

Types of Cyber Crimes (List)

1. Hacking
2. Phishing
3. Identity Theft
4. Password Attack
5. Malware Attack
6. Virus
7. Worm
8. Trojan Horse

9. Ransomware
10. Spyware
11. Adware
12. Keylogger Attack
13. Botnet Attack
14. Rootkit Attack
15. Logic Bomb
16. Email Spoofing
17. SMS Spoofing
18. Website Defacement
19. Website Hijacking
20. SQL Injection Attack
21. Cross-Site Scripting (XSS)
22. Denial of Service (DoS) Attack
23. Distributed Denial of Service (DDoS) Attack
24. Man-in-the-Middle (MITM) Attack
25. Session Hijacking
26. Sniffing
27. Social Engineering Attack
28. Cyber Stalking
29. Cyber Bullying
30. Cyber Harassment
31. Online Banking Fraud
32. UPI Fraud
33. Credit/Debit Card Fraud
34. ATM Fraud
35. Internet Banking Fraud
36. E-commerce Fraud
37. SIM Swap Fraud
38. OTP Fraud
39. QR Code Scam
40. Fake Mobile Applications
41. Fake Websites
42. Fake Job Scam
43. Online Investment Scam
44. Cryptocurrency Fraud
45. Identity Fraud
46. Data Theft
47. Data Breach
48. Intellectual Property (IP) Theft
49. Software Piracy
50. Digital Piracy
51. Copyright Infringement
52. Cyber Espionage
53. Cyber Terrorism
54. Cyber Warfare
55. Child Pornography / Child Sexual Exploitation
56. Online Human Trafficking
57. Revenge Porn
58. Deepfake Crime
59. AI-Enabled Cyber Crime

60. IoT-Based Cyber Attack
61. Cloud Security Attack
62. Mobile Device Attack
63. Insider Threat
64. Business Email Compromise (BEC)
65. Supply Chain Attack
66. DNS Spoofing
67. DNS Cache Poisoning
68. Zero-Day Attack
69. Cryptojacking
70. Fake News and Misinformation Campaigns

Classification of Cyber Crimes

Cyber crimes are commonly classified based on **the target or victim** of the attack. This classification helps in understanding the nature of the crime, its impact, and the applicable legal provisions.

Classification of Cyber Crimes

1. Crimes Against Individuals

Definition

These are cyber crimes that target an individual person. The main objective is to steal personal information, harass, threaten, defame, or cause financial loss.

Examples:

- Identity Theft
- Phishing
- Cyber Stalking
- Cyber Bullying
- Email Spoofing
- Online Banking Fraud
- Credit Card Fraud
- Social Media Account Hacking
- OTP Fraud

2. Crimes Against Property

Definition These crimes target digital assets, computer systems, software, databases, and intellectual property.

Examples:

- Data Theft
- Software Piracy
- Malware Attack
- Virus
- Worm
- Trojan Horse
- Ransomware
- Website Defacement
- Intellectual Property Theft

3. Crimes Against Organizations

Definition (English):
These crimes target companies, educational institutions, hospitals, banks, or other organizations to steal confidential information or disrupt business operations.

Examples:

- Database Hacking
- Business Email Compromise (BEC)
- Insider Attack
- DDoS Attack
- Corporate Espionage
- Data Breach
- Supply Chain Attack

4. Crimes Against Society

Definition

These cyber crimes affect society as a whole by spreading illegal content, misinformation, or activities that disturb public order.

Examples:

- Fake News
- Online Gambling
- Hate Speech
- Child Sexual Exploitation Material (CSEM)
- Drug Trafficking through the Internet

- Human Trafficking
- Illegal Online Content

5. Crimes Against Government

Definition These crimes target government organizations, military systems, public infrastructure, or national security.

Examples:

- Cyber Terrorism
- Cyber Espionage
- Government Website Hacking
- Attack on Defence Networks
- Power Grid Attack
- Election System Attack
- Critical Infrastructure Attack

Summary Table

Classification	Target	Main Objective	Examples
Crimes Against Individuals	Individual Person	Identity theft, harassment, financial fraud	Phishing, Cyber Stalking, OTP Fraud
Crimes Against Property	Data, Software, Computer Systems	Damage or steal digital assets	Virus, Ransomware, Data Theft
Crimes Against Organizations	Companies, Banks, Universities	Disrupt operations or steal confidential data	DDoS, Database Hacking, BEC
Crimes Against Society	Society/Public	Spread illegal content or create social disorder	Fake News, Online Gambling, Hate Speech
Crimes Against Government	Government & National Infrastructure	Threaten national security	Cyber Terrorism, Cyber Espionage

Easy Memory Trick

Remember the order:

I → P → O → S → G

- **I** = Individuals
- **P** = Property
- **O** = Organizations

- **S** = Society
- **G** = Government

Distinction between Cyber Crimes and Conventional Crime

Introduction

Crime is generally defined as an unlawful act punishable by law. Crimes can be broadly classified into **Conventional Crimes (Traditional Crimes)** and **Cyber Crimes**. Conventional crimes have existed for centuries and are committed in the physical world, whereas cyber crimes emerged with the development of computers, the Internet, and digital technologies.

With the rapid growth of information technology, criminals have shifted from traditional methods to digital methods, making cyber crimes more complex, borderless, and difficult to investigate.

Difference between Cyber Crime and Conventional Crime

Parameter	Cyber Crime	Conventional Crime
Definition	Crime committed using computers, digital devices, or the Internet.	Crime committed in the physical world without using digital technology.
Medium	Digital (Internet, Computer, Mobile, Network)	Physical (Person, Weapon, Vehicle, etc.)
Place of Crime	Can be committed from anywhere in the world.	Usually committed at a physical location.
Criminal's Presence	Physical presence is generally not required.	Physical presence is usually required.

Target	Computer systems, networks, data, websites, digital users.	People, property, vehicles, buildings, etc.
Evidence	Digital Evidence (Logs, Emails, IP Address, Hard Disk, CCTV Data).	Physical Evidence (Fingerprints, Weapons, Documents, Witnesses).
Investigation	Requires Digital Forensics and Cyber Experts.	Requires Police Investigation and Physical Evidence Collection.
Speed	Can be completed within seconds or minutes.	Usually takes more time.
Geographical Boundary	Borderless (Global).	Mostly Local or National.
Identity of Criminal	Criminal can hide identity using VPN, Proxy, Fake Accounts, Encryption.	Criminal is comparatively easier to identify.
Financial Loss	Can affect millions of users instantly.	Usually limited to one victim or one location.
Examples	Phishing, Hacking, Ransomware, UPI Fraud, Identity Theft.	Theft, Murder, Robbery, Burglary, Assault, Kidnapping.
Applicable Laws (India)	Information Technology Act, 2000 + IPC/BNS (where applicable).	IPC/BNS and other criminal laws.

Example 1: Cyber Crime

A hacker creates a fake banking website that looks exactly like the official website. A customer enters their username, password, and OTP. The hacker steals the money from the customer's account.

This is a Cyber Crime.

Example 2: Conventional Crime

A thief enters a house at night and steals jewellery and cash.

This is a Conventional Crime.

Real-Life Comparison

Situation	Type of Crime
Someone steals your ATM PIN using a fake website.	Cyber Crime
Someone steals your ATM card from your pocket.	Conventional Crime
A hacker locks your files and demands ransom.	Cyber Crime
A thief steals your laptop from your office.	Conventional Crime

Someone hacks your email account.	Cyber Crime
Someone breaks into your house and steals documents.	Conventional Crime

Memory Trick

Cyber Crime = "Digital World"

- Internet
- Computer
- Data
- Network
- Online

Conventional Crime = "Physical World"

- Person
- Property
- Weapon
- Building
- Physical Evidence

Need of Cyber Law

Introduction

English

The rapid growth of computers, the Internet, smartphones, cloud computing, digital payments, and e-commerce has transformed the way people live and work. Today, millions of people use online services for banking, education, healthcare, shopping, communication, and government services. Along with these benefits, cyber crimes such as hacking, phishing, identity theft, ransomware, and online fraud have also increased significantly.

To regulate activities in cyberspace, protect users, secure digital transactions, and punish cyber criminals, **Cyber Law** has become essential. It provides a legal framework for the safe and lawful use of digital technologies.

Need of Cyber Law (Why is Cyber Law Important?)

1. To Prevent Cyber Crimes

Cyber Law helps prevent crimes such as hacking, phishing, ransomware, identity theft, cyber stalking, and online banking fraud by defining offences and prescribing punishments.

Example: A hacker who steals banking credentials can be prosecuted under the Information Technology Act, 2000.

2. To Provide Legal Recognition to Electronic Records

Cyber Law recognizes electronic records, emails, electronic contracts, and digital documents as legally valid.

Example: Online university admission forms and electronic invoices are legally accepted.

3. To Support E-Commerce

Cyber Law builds trust in online shopping by protecting customers and businesses from fraud.

Example: Amazon, Flipkart, and online marketplaces rely on secure digital transactions.

4. To Secure Online Banking and Digital Payments

Cyber Law protects electronic banking systems, credit/debit card transactions, UPI payments, and mobile banking services.

Example: Action can be taken against UPI fraud and unauthorized online transactions.

5. To Protect Privacy and Personal Data

Cyber Law safeguards personal information from unauthorized access, misuse, or disclosure.

Example: Protecting Aadhaar, PAN, passport details, and medical records from unauthorized use.

6. To Protect Intellectual Property Rights (IPR)

Cyber Law protects software, digital content, research papers, music, movies, and other intellectual property from piracy and unauthorized copying.

Example: Illegal downloading or copying of licensed software.

7. To Regulate Digital Communication

Cyber Law regulates emails, social media platforms, messaging applications, and electronic communications.

Example: Legal action against fake social media accounts spreading misinformation.

8. To Protect National Security

English

Cyber Law helps protect critical infrastructure such as defence systems, power grids, government databases, and communication networks from cyber attacks.

Example: Preventing cyber attacks on government websites and defence networks.

9. To Ensure Digital Evidence is Legally Accepted

Cyber Law recognizes digital evidence such as emails, CCTV recordings, server logs, and electronic documents in courts.

Example: Email records used as evidence in cyber fraud cases.

10. To Promote Safe Digital Society

Cyber Law encourages responsible use of technology and creates a secure environment for citizens, businesses, and governments.

Summary Table

Need of Cyber Law	Purpose	Example
Prevent Cyber Crimes	Punish cyber offenders	Hacking, Phishing

Legal Recognition	Validate electronic records	E-contracts, Digital Signatures
Secure E-Commerce	Safe online business	Online shopping
Secure Banking	Protect digital payments	UPI, Net Banking
Protect Privacy	Safeguard personal data	Aadhaar, PAN
Protect IPR	Prevent software piracy	Licensed software
Regulate Communication	Control misuse of social media	Fake accounts
Protect National Security	Secure government systems	Defence networks
Accept Digital Evidence	Support investigations	Email, CCTV, Server logs
Promote Digital Society	Safe use of technology	E-Governance

Need of Cyber Law in India

In India, the increasing use of digital payments, online banking, UPI, e-governance, and e-commerce has made Cyber Law extremely important. The **Information Technology Act, 2000** provides the legal framework for electronic records, digital signatures, cyber offences, and penalties. Institutions such as **CERT-In** and the **National Cyber Crime Reporting Portal** strengthen India's cyber security ecosystem.

Cyber Criminals

Introduction

A **Cyber Criminal** is a person, group, or organization that uses computers, digital devices, networks, or the Internet to perform illegal activities. Their objective may be to steal money, confidential information, intellectual property, or to damage computer systems and networks. With the advancement of technology, cyber criminals have become more organized and use sophisticated tools such as malware, ransomware, phishing, artificial intelligence (AI), and social engineering.

Definition

A Cyber Criminal is an individual or group that intentionally uses computers, networks, or the Internet to commit illegal activities for financial gain, personal benefit, revenge, political motives, or other malicious purposes.

Characteristics of Cyber Criminals

- Technically skilled in computers and networks.
- Uses anonymity to hide identity.
- Operates from any location in the world.
- Targets individuals, organizations, or governments.
- Uses advanced cyber attack techniques.

Types of Cyber Criminals

1. Black Hat Hackers

Black Hat Hackers illegally access computer systems to steal data, spread malware, or cause damage.

Example: A hacker steals customers' credit card information from an online shopping website.

2. White Hat Hackers

White Hat Hackers are ethical hackers who test the security of systems with permission to identify vulnerabilities.

Example: A bank hires an ethical hacker to test its online banking security.

3. Gray Hat Hackers

Gray Hat Hackers access systems without permission but usually do not intend to cause harm. They often report vulnerabilities after discovering them.

Example: A person discovers a security flaw in a website and informs the company.

4. Script Kiddies

Script Kiddies have limited technical knowledge and use ready-made hacking tools developed by others.

Example: Using a free DDoS tool downloaded from the Internet.

5. Insider Threats

Insider Threats are employees or authorized users who misuse their access to steal or damage organizational data.

Example: An employee copies confidential customer data before leaving the company.

6. Organized Cyber Crime Groups

These are professional criminal organizations that conduct large-scale cyber attacks for financial gain.

Example: A ransomware gang attacking multiple companies across different countries.

7. Cyber Terrorists

Cyber Terrorists attack government systems, military networks, power grids, or critical infrastructure to create fear or threaten national security.

Example: Attacking a country's power grid or defence communication network.

8. State-Sponsored Hackers

These hackers work on behalf of a government to conduct cyber espionage or attack another country's digital infrastructure.

Example: A cyber espionage attack targeting another country's defence network.

Motives of Cyber Criminals

Motive	Description	Example
Financial Gain	Stealing money	UPI Fraud
Data Theft	Stealing confidential data	Customer Database Theft
Revenge	Personal revenge	Hacking ex-employer's account
Fame	Showing hacking skills	Website Defacement
Political Motive	Influencing politics	Election-related cyber attack

Cyber Espionage	Stealing secret information	Government data theft
Terrorism	Creating fear	Attack on critical infrastructure

Common Tools Used by Cyber Criminals

- Malware
- Virus
- Worm
- Trojan Horse
- Ransomware
- Spyware
- Keylogger
- Botnet
- Phishing Kits
- Password Cracking Tools
- Social Engineering Techniques

How to Protect Yourself from Cyber Criminals

- Use strong and unique passwords.
- Enable Multi-Factor Authentication (MFA).
- Keep software updated.
- Install antivirus software.
- Never share OTP or passwords.
- Avoid suspicious emails and links.
- Backup important data regularly.

Summary Table

Type of Cyber Criminal	**Main Objective**	**Example**
Black Hat Hacker	Illegal hacking	Credit card data theft
White Hat Hacker	Improve security	Penetration testing
Gray Hat Hacker	Find vulnerabilities	Security flaw reporting
Script Kiddie	Use ready-made hacking tools	Basic DDoS attack
Insider Threat	Misuse authorized access	Employee stealing company data
Organized Crime Group	Financial gain	Ransomware gang

Cyber Terrorist	Threaten national security	Attack on government systems
State-Sponsored Hacker	Cyber espionage	Defence network attack

Cyber Crime in

Introduction

English

India is one of the fastest-growing digital economies in the world. The rapid growth of the Internet, smartphones, digital payments (UPI), e-commerce, cloud computing, and e-governance has transformed the way people live and work. However, this digital transformation has also led to a significant increase in cyber crimes such as phishing, hacking, identity theft, ransomware, online banking fraud, and social media scams.

To combat these crimes, the Government of India has introduced several laws, institutions, and initiatives, including the **Information Technology (IT) Act, 2000**, **CERT-In**, and the **National Cyber Crime Reporting Portal**.

Reasons for the Growth of Cyber Crime in India

- Rapid growth of Internet users.
- Increased use of smartphones.
- Growth of UPI and digital payment systems.
- Expansion of e-commerce and online banking.
- Lack of cyber security awareness.
- Weak passwords and poor security practices.
- Increased use of social media.

Common Cyber Crimes in India

1. Phishing
2. Hacking
3. Identity Theft
4. Online Banking Fraud
5. UPI Fraud
6. Credit/Debit Card Fraud
7. Cyber Stalking

8. Cyber Bullying
9. Ransomware Attack
10. Social Media Fraud Fake Job Scam
11. OTP Fraud

Major Cyber Crime Incidents in India

1. Cosmos Bank Cyber Attack (2018)

- **English:** Hackers compromised the bank's systems and fraudulently withdrew large amounts of money through ATM and SWIFT transactions.

2. Kudankulam Nuclear Power Plant Malware Incident (2019)

- **English:** Malware was detected in the administrative network of the Kudankulam Nuclear Power Plant.

3. Air India Data Breach (2021)

- **English:** Personal information of millions of passengers was compromised in a major data breach.

4. AIIMS Delhi Ransomware Attack (2022)

- **English:** The AIIMS Delhi hospital's servers were infected with ransomware, disrupting hospital services and access to patient records.

Cyber Laws in India

Information Technology (IT) Act, 2000

The IT Act, 2000 is the primary law dealing with cyber crimes and electronic transactions in India.

Objectives

- Legal recognition of electronic records.
- Legal recognition of digital signatures.
- Prevention of cyber crimes.
- Punishment for cyber offenders.
- Promotion of secure e-commerce.

Government Initiatives

1. CERT-In

Indian Computer Emergency Response Team

Functions

- Responds to cyber security incidents.
- Issues security alerts and advisories.
- Coordinates cyber incident response.

2. National Cyber Crime Reporting Portal

An online platform where citizens can report cyber crimes, especially financial fraud and crimes against women and children.

3. Cyber Swachhta Kendra

Provides free tools and guidance to detect and remove malware from computers and mobile devices.

4. Indian Cyber Crime Coordination Centre (I4C)

Coordinates cyber crime investigation, intelligence sharing, training, and capacity building among law enforcement agencies.

How to Prevent Cyber Crime

- Use strong passwords.
- Enable Multi-Factor Authentication (MFA).
- Do not share OTP or PIN.
- Keep software updated.
- Install antivirus software.
- Verify websites before making payments.
- Avoid clicking unknown links.
- Regularly back up important data.

Summary Table

Aspect	Details
Main Law	Information Technology Act, 2000
Major Crimes	Phishing, Hacking, UPI Fraud, Identity Theft, Ransomware
Government Agencies	CERT-In, I4C
Reporting Platform	National Cyber Crime Reporting Portal
Malware Removal	Cyber Swachhta Kendra
Prevention	Strong Passwords, MFA, Awareness, Software Updates

Unit-2

Information Technology Act 2000 Introduction to IT Act 2000:-

Objective of the IT Act 2000,structure of the Act, features of IT Act 2000,Important chapters of IT Act, summary of the Act, Amendment bill 2008

Information Technology Act, 2000 (IT Act, 2000)

Introduction to Information Technology Act, 2000

The **Information Technology (IT) Act, 2000** is India's first cyber law. It was enacted by the Government of India and came into force on **17 October 2000**. The Act provides a legal framework for electronic records, digital signatures, electronic transactions, e-commerce, and cyber crimes.

The main purpose of the IT Act is to promote secure electronic communication, facilitate e-commerce and e-governance, and prevent cyber crimes.

The Act is based on the **UNCITRAL Model Law on Electronic Commerce (1996)** adopted by the United Nations to encourage uniform electronic commerce laws across countries.

Objectives of IT Act, 2000

- To provide legal recognition to electronic records.
- To recognize digital signatures.
- To promote e-commerce and e-governance.
- To prevent cyber crimes.
- To ensure secure electronic transactions.
- To protect users in cyberspace.

Key Features of IT Act, 2000

- Legal recognition of **Electronic Records**.
- Recognition of **Digital Signatures**.
- Regulation of **Electronic Transactions**.
- Punishment for **Cyber Crimes**.
- Establishment of **Certifying Authorities**.
- Protection of **Digital Information**.
- Promotion of **E-Commerce and E-Governance**.

Importance of IT Act, 2000

- Provides legal protection for online activities.
- Helps control cyber crimes.
- Supports digital banking and online business.
- Protects electronic data and privacy.
- Builds trust in digital services.

Examples

Example 1

A customer purchases a product through an online shopping website. The electronic invoice and payment receipt are legally valid under the IT Act, 2000.

Example 2

If a hacker steals confidential data from a company's computer system, legal action can be taken under the IT Act, 2000.

Summary

Aspect	Details
Name	Information Technology Act, 2000
Country	India
Implemented On	**17 October 2000**
Purpose	To regulate electronic records, digital signatures, e-commerce, and cyber crimes
Based On	UNCITRAL Model Law on Electronic Commerce (1996)
Main Focus	Cyber Law, Electronic Transactions, Digital Security

Objectives of the Information Technology (IT) Act, 2000 Introduction

The **Information Technology Act, 2000** was enacted by the Government of India to provide a legal framework for electronic transactions, digital communication, and cyber security. The Act aims to facilitate the growth of e-commerce, e-governance, and secure online services while preventing cyber crimes.

Objectives of the IT Act, 2000

1. Legal Recognition of Electronic Records

To provide legal recognition to electronic records so that digital documents are accepted as valid evidence.

Example: E-mails, electronic invoices, and online application forms.

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2. Legal Recognition of Digital Signatures

To recognize digital signatures as a valid method of authentication for electronic documents.

Example: Signing an online government tender using a Digital Signature Certificate (DSC).

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3. Promote E-Commerce

To encourage electronic commerce by providing legal protection to online business transactions.

Example: Shopping through Amazon or Flipkart.

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4. Promote E-Governance

To enable government departments to provide services electronically.

Example: Online income tax filing and passport application.

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5. Prevent Cyber Crimes

To define cyber offences and prescribe punishments for cyber criminals.

Example: Hacking, phishing, identity theft, and ransomware attacks.

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6. Secure Electronic Transactions

To ensure that online transactions are secure, reliable, and trustworthy.

Example: UPI payments and Internet banking.

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7. Protect Data and Privacy

To protect electronic data and personal information from unauthorized access and misuse.

Example: Protecting Aadhaar and banking details.

8. Establish Certifying Authorities

To regulate Certifying Authorities that issue Digital Signature Certificates (DSCs).

9. Encourage Secure Digital Communication

To ensure secure communication through electronic means.

Example: Secure e-mail communication and encrypted messaging.

10. Build Trust in the Digital Economy

To create confidence among citizens and businesses in using digital technologies and online services.

Summary Table

Objective	Purpose	Example
Legal Recognition of Electronic Records	Validate digital documents	E-mail, E-Invoice
Legal Recognition of Digital Signatures	Authenticate electronic documents	Digital Signature Certificate (DSC)
Promote E-Commerce	Secure online business	Online shopping
Promote E-Governance	Digital government services	Online passport application

Objective	Purpose	Example
Prevent Cyber Crimes	Define offences and punishments	Hacking, Phishing
Secure Electronic Transactions	Protect online payments	UPI, Net Banking
Protect Data and Privacy	Safeguard personal information	Aadhaar, PAN
Establish Certifying Authorities	Regulate issuance of DSCs	Licensed Certifying Authorities
Secure Digital Communication	Safe exchange of information	Encrypted e-mails
Build Trust in Digital Economy	Increase confidence in digital services	Digital banking

Structure of the Information Technology (IT) Act, 2000

Introduction

The **Information Technology (IT) Act, 2000** is India's primary cyber law. It provides the legal framework for electronic records, digital signatures, e-commerce, e-governance, and cyber crimes. The Act is systematically divided into **Chapters and Sections**, where each chapter deals with a specific aspect of cyber law.

Structure of the IT Act, 2000

The Act is organized into **13 Chapters and 94 Sections** (after amendments).

Chapter	Title	Purpose
Chapter I	Preliminary	Defines the title, extent, commencement, and important definitions used in the Act.
Chapter II	Digital Signature	Provides legal recognition and rules relating to Digital Signatures and authentication of electronic records.
Chapter III	Electronic Governance	Recognizes electronic records, electronic filing, storage, and communication with government departments.
Chapter IV	Attribution, Acknowledgement and Dispatch of Electronic Records	Explains how electronic records are sent, received, and legally recognized.
Chapter V	Secure Electronic Records and Secure Digital Signatures	Describes secure electronic records and secure digital signatures.
Chapter VI	Regulation of Certifying Authorities	Provides rules for licensing and regulating Certifying Authorities that issue Digital Signature Certificates (DSCs).
Chapter VII	Digital Signature Certificates	Explains the issue, suspension, and revocation of Digital Signature Certificates.
Chapter VIII	Duties of Subscribers	Describes the responsibilities of users (subscribers) of Digital Signature Certificates.
Chapter IX	Penalties and Adjudication	Specifies penalties, compensation, and appointment of adjudicating officers for cyber offences.
Chapter X	Cyber Appellate Tribunal *(now merged with TDSAT)*	Provides a mechanism for appeals against decisions related to cyber disputes.
Chapter	Offences	Defines cyber offences such as hacking,

Chapter	Title	Purpose
XI		identity theft, cheating by personation, and publishing obscene content, along with punishments.
Chapter XII	Intermediary Liability	Specifies the responsibilities and legal protection of intermediaries such as Internet Service Providers (ISPs) and social media platforms under certain conditions.
Chapter XIII	Miscellaneous	Contains miscellaneous provisions related to the implementation and administration of the Act.

Simple Flow of the IT Act

IT Act, 2000
- Preliminary
- Digital Signatures
- Electronic Governance
- Electronic Records
- Secure Electronic Records
- Certifying Authorities
- Digital Signature Certificates
- Duties of Subscribers
- Penalties & Adjudication
- Appeals
- Cyber Offences
- Intermediary Liability
- Miscellaneous Provisions

Importance of the Structure

- Organizes the law into logical chapters.
- Makes legal provisions easy to understand.
- Covers electronic transactions, digital signatures, cyber offences, and penalties in a systematic way.
- Helps students, lawyers, and investigators quickly locate relevant provisions.

Features of the Information Technology (IT) Act, 2000

Introduction

The **Information Technology (IT) Act, 2000** is India's first cyber law. It provides a legal framework for electronic records, digital signatures, electronic commerce, electronic governance, and cyber security. The Act promotes secure electronic transactions and prescribes penalties for cyber offences.

Features of IT Act, 2000

1. Legal Recognition of Electronic Records

The Act gives legal recognition to electronic records, making digital documents valid in legal and business transactions.

Example: E-mails, electronic invoices, and online application forms.

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2. Legal Recognition of Digital Signatures

The Act recognizes Digital Signatures as a valid method of authenticating electronic documents.

Example: Signing government tenders using a Digital Signature Certificate (DSC).

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3. Promotion of E-Commerce

The Act provides legal support for online business and electronic transactions.

Example: Online shopping and digital payments.

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4. Support for E-Governance

The Act allows government departments to provide services electronically.

Example: Online passport application and income tax filing.

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5. Recognition of Electronic Contracts

Electronic contracts (e-contracts) made through electronic communication are legally valid.

Example: Accepting the "Terms and Conditions" while purchasing software online.

6. Regulation of Certifying Authorities

The Act provides for licensing and regulation of Certifying Authorities that issue Digital Signature Certificates.

7. Defines Cyber Offences and Penalties

The Act defines various cyber offences and prescribes penalties and punishments.

Examples: Hacking, identity theft, phishing, and cyber fraud.

8. Protection of Data and Privacy

The Act provides legal protection against unauthorized access, misuse, and disclosure of electronic data.

9. Recognition of Electronic Evidence

Electronic records and digital evidence can be used in legal proceedings.

Example: E-mails, CCTV recordings, server logs.

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10. Promotes Secure Digital Environment

The Act encourages secure use of computers, networks, and the Internet by establishing legal responsibilities and penalties.

Summary Table

Feature	Purpose	Example
Electronic Records	Legal recognition of digital documents	E-mail, E-Invoice
Digital Signatures	Authentication of documents	DSC
E-Commerce	Secure online business	Online shopping
E-Governance	Digital government services	Online tax filing
Electronic Contracts	Legal validity of online agreements	Software license agreement
Certifying Authorities	Issue Digital Signature Certificates	Licensed CA
Cyber Offences	Define crimes and punishments	Hacking, Phishing
Data Protection	Protect personal information	Aadhaar, PAN
Electronic Evidence	Accept digital evidence in court	E-mails, CCTV
Secure Digital Environment	Safe use of technology	Online banking

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Summary of the Information Technology (IT) Act, 2000

Introduction

The **Information Technology (IT) Act, 2000** is India's first and primary cyber law. It was enacted by the Government of India and came into force on **17 October 2000**. The Act provides a legal framework for electronic records, digital

signatures, electronic transactions, e-commerce, e-governance, and cyber security. It also defines cyber offences and prescribes penalties for offenders.

Key Highlights

Aspect	Description
Act Name	Information Technology Act, 2000
Enforced On	17 October 2000
Purpose	Legal framework for electronic transactions and cyber law
Main Focus	Electronic Records, Digital Signatures, E-Commerce, E-Governance
Cyber Crimes Covered	Hacking, Identity Theft, Phishing, Cyber Fraud, Data Theft
Legal Recognition	Electronic Records and Digital Signatures
Evidence	Electronic records are admissible in courts
Importance	Promotes secure digital transactions and cyber security

Information Technology (Amendment) Act, 2008

Introduction

The **Information Technology (Amendment) Act, 2008** was enacted to update the **Information Technology Act, 2000** in response to the rapid growth of the Internet, e-commerce, online banking, social media, and emerging cyber crimes. The amendment was **passed by the Indian Parliament in December 2008** and **came into force on 27 October 2009**.

It introduced new cyber offences, strengthened data protection provisions, recognized electronic signatures, and increased the responsibility of intermediaries such as Internet Service Providers (ISPs) and online platforms.

Objectives of the IT (Amendment) Act, 2008

English

- To strengthen cyber security laws.
- To address new forms of cyber crimes.
- To protect sensitive personal data.
- To recognize electronic signatures.
- To regulate intermediaries.
- To improve investigation and prosecution of cyber offences.

Major Amendments in the IT Act, 2008

1. Electronic Signature

- **English:** Introduced the concept of **Electronic Signature** in addition to Digital Signature.

2. Data Protection

- **English:** Introduced provisions to protect sensitive personal information.

3. New Cyber Offences

The Amendment introduced several new offences such as:

- Identity Theft (Section 66C)
- Cheating by Personation using Computer Resources (Section 66D)
- Violation of Privacy (Section 66E)
- Cyber Terrorism (Section 66F)

4. Intermediary Liability (Section 79)

- **English:** Intermediaries are given protection from liability if they follow legal requirements and exercise due diligence.

5. Protection of Critical Information Infrastructure

- **English:** Strengthened protection for critical computer systems and national infrastructure.

6. Government Powers

- **English:** The Government was given powers to intercept, monitor, or decrypt information under specific legal conditions.

Important New Sections Introduced

Section	Topic	Description
66C	Identity Theft	Punishment for misuse of passwords, digital signatures, and electronic identities.
66D	Cheating by Personation	Punishment for online fraud using fake identity.
66E	Violation of Privacy	Punishment for capturing or sharing private images without consent.
66F	Cyber Terrorism	Punishment for cyber attacks threatening national security.
67A	Sexually Explicit Material	Punishment for publishing or transmitting sexually explicit content online.
67B	Child Sexual Content	Punishment for child sexual abuse material online.
72A	Disclosure of Information	Punishment for unlawful disclosure of personal information.

Importance of the Amendment

- Updated the IT Act to address modern cyber crimes.
- Improved protection of personal data and privacy.
- Increased accountability of online service providers.

- Enhanced national cyber security.

Summary Table

Feature	IT Act, 2000	IT (Amendment) Act, 2008
Digital Signature	✔️	✔️
Electronic Signature	✘	✔️
Identity Theft Law	Limited	✔️ Section 66C
Cyber Terrorism	Not defined	✔️ Section 66F
Data Protection	Basic	Strengthened
Intermediary Liability	Limited	Expanded (Section 79)
Privacy Protection	Limited	Improved